

CIE

COPYRIGHT INFORMATION EXTRACTION FORM

TITLE: **Synaptix: Autonomous Multi-Agent Framework for Semantic Codebase Navigation and Architectural Mapping**

SCHOOL: Chitkara University, Rajpura, Punjab

**DETAIL OF THE STUDENT/MENTOR WHO UPLOADED THE COPYRIGHT ON GOVT PORTAL NEED TO
SHARE LOGIN AND PASSWORD WITH MENTORS**

Name: Vanshaj Gupta

Roll No/ Employee Code: 2210990941

ALL MEMBERS DETAILS INCLUDING SUPERVISOR

NAME	EMP CODE/ROLL	ADDRESS WITH EMAIL AND MOBILE NO.
Student 1: Vanshaj Gupta	2210990941	vanshaj941.be22@chitkara.edu.in , #26, Satsang Vihar, Ambala Cantt – 133001, +91 – 85709 30737
Student 2: Arjun Pathak	2210991340	arjun1340.be22@chitkara.edu.in , Ward No. 9, Mehta Colony, Una, Himachal Pradesh -174303, +91 – 82191 45181
Student 3: Rohan Kapoor	2210990738	rohan738.be22@chitkara.edu.in , #612, Sunny Enclave, Devigarh Road, Patiala - 147111, +91 – 79735 31450
Student 4: Arnav Gupta	2210990161	arnav161.be22@chitkara.edu.in , #149, ward-15, bagh sodhian, Kurali, Punjab - 147111, +91 - 86998 21510
Mentor: Dr. Rajat Takkar		

ABSTRACT

1.1 Problem Background

Modern software development suffers from increasing complexity, where large repositories often lack up-to-date documentation. Developers joining new teams face a steep learning curve, typically spending excessive time manually tracing logic flow across thousands of files rather than contributing effective code. Traditional tools like static analysis provide syntax checking but fail to offer semantic understanding or explain *how* components interact dynamically. This "knowledge gap" slows down onboarding, increases technical debt, and makes maintaining legacy systems high-risk, as the original architectural intent is often lost when key personnel leave the organization. Existing AI solutions often treat code as generic text, leading to "hallucinated" explanations that do not reflect the actual file dependencies.

1.2 Objective of the Project

The primary objective is to develop **Synaptix**, an agentic AI system that autonomously constructs a "Mental Map" of any software repository. By automating context extraction and dependency tracing, the project aims to reduce developer onboarding time from weeks to minutes, providing real-time, self-correcting architectural intelligence.

1.3 Proposed Solution / System Overview

Synaptix creates a Directed Acyclic Graph (DAG) of specialized AI agents designed to mimic a senior engineer's code exploration process. Unlike passive summarizers, the system employs a proprietary "Context-Climbing" algorithm. It identifies entry points (e.g., `main.py`) and recursively traverses import statements to map the application's skeleton. The architecture comprises a Discovery Agent for file system scanning, a Semantic Indexer for vectorizing code chunks, and a Relationship Resolver that links function calls across files. These components synthesize a holistic understanding, ensuring documentation evolves synchronously with the source code.

1.4 Key Features or Functionality

Key features include Dynamic Logical Tracing, which generates real-time Mermaid.js flowcharts to visualize data paths (e.g., API → Controller → Database). The system features a unique **"Critic Node"** mechanism that verifies AI-generated explanations against raw code to minimize hallucinations. Additionally, Synaptix performs Architectural Gap Analysis, automatically identifying missing error handlers or broken dependency chains, effectively transforming static repositories into interactive, queryable knowledge bases.

1.5 Technology / Method Used

The system is built on Python and LangGraph for stateful multi-agent orchestration. It utilizes ChromaDB for local, privacy-first vector storage and supports a hybrid deployment model: running entirely offline via Ollama for zero-cost local inference, or scaling to cloud-based LLMs for enterprise performance without architectural refactoring.

1.6 Expected Outcome & Benefits

The project delivers a dramatic reduction in "Time-to-Understanding" (TTU) for developers, enabling zero-day onboarding for complex repositories. It facilitates safer maintenance of legacy systems by exposing hidden dependencies and logic flaws. Furthermore, for open-source communities, it automates the identification of "Good First Issues," making large projects more accessible. Ultimately, Synaptix transforms code into a self-documenting asset, bridging the gap between code implementation and architectural intent.

1.7 Original Contribution Statement

The original contribution lies in the proprietary State Schema and Routing Logic that enables autonomous "deep-dive" versus "summary" decision-making within the agentic workflow, specifically the integration of the self-correcting "Critic Node" for verification.